

IIITH- ALL BRANCHES ENGINEERING MATHEMATICS



Lecture No. 01

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Recap of previous lecture



Topic

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Topics to be Covered



Topic

Linear Algebra



The IIIT Hyderabad Postgraduate Entrance Examination (PGEE) for 2026 consists of two, 90-minute sections: General Aptitude (50% weightage, includes basic math, logic, and computing) and a Subject-Specific technical paper (50% weightage, covering core CSE or ECE topics). The technical syllabus is aligned with GATE but often omits some topics like compiler design, focusing on core fundamentals.  IIIT Hyderabad +3

PGEE Exam Structure

- **Total Time:** 3 hours (180 minutes).
- **Sections:** Two, 90-minute sessions.
 - **Part 1: General Aptitude & Mathematics:** Basic Aptitude, Logical Reasoning, Basic Computer Knowledge, and Mathematics (Calculus, Probability, Algebra).
 - **Part 2: Subject Paper:** Core CS or ECE subjects.
- **Negative Marking:** Yes, 25% for wrong answers.  IIIT Hyderabad +4

✓ Linear Algebra: Matrices, Rank, Determinants and its properties, Eigenvalues and Eigenvectors.

✓ Calculus: Integration, Differentiation, Limits, Continuity and Differentiability, Convergence and Divergence of Series, Maxima, and Minima, Mean value theorem, Partial Derivatives, Ordinary Differential Equations and Applications, Initial and Boundary Value Problems.

✓ Probability Theory

Conditional Probability | Bayes Theorem | Random variable | PDF and CDF | Moment Generating Function | Theoretical Distribution (Binomial, Poisson, Normal, Uniform and Hypergeometric)

- ① Determinants
- ② Matrices
- ③ Types of Matrices

Det of $3 \times 3 \rightarrow |A| = \begin{vmatrix} + & - & + \\ - & + & - \\ + & - & + \end{vmatrix}$

9 $|A| = \begin{vmatrix} 2 & -3 & 1 \\ 4 & 2 & 0 \\ -1 & 3 & 5 \end{vmatrix} = ? = -4 \begin{vmatrix} -3 & 1 \\ 3 & 5 \end{vmatrix} + 2 \begin{vmatrix} 2 & 1 \\ -1 & 5 \end{vmatrix} - (0) \begin{vmatrix} 2 & 1 \\ -1 & 5 \end{vmatrix} ?$

$= -4[-15-3] + 2[10+1] - 0$

$= +72 + 22 = \textcircled{94} \underline{\underline{Ans}}$

(11) SARRUS method $\rightarrow$

$$\begin{array}{ccccc} 2 & -3 & 1 & 2 & -3 \\ 4 & 2 & 0 & 4 & 2 \\ -1 & 3 & 5 & -1 & 3 \end{array}$$

80 $|A| = (20+0+12) - (-2+0-60)$

$= 32 + 62 = \textcircled{94} \underline{\underline{Ans}}$

Elementary Operations →



E-Row Operations

- (i) $R_i \leftrightarrow R_j$
- (ii) $R_i \rightarrow k R_i$
- (iii) $R_i \rightarrow R_i + k R_j$

$$R_3 \rightarrow R_3 + 5R_2 \quad (\text{E-operation of 3rd type})$$

$$R_3 \rightarrow 5R_3 + R_2 \quad (\text{it is not an E-operation, it is Mixed operation})$$

$$R_3 \rightarrow 5R_3 \text{ \& then } R_3 \rightarrow R_3 + R_2$$

E-Column Operations

- (i) $C_i \leftrightarrow C_j$
- (ii) $C_i \rightarrow k C_i$
- (iii) $C_i \rightarrow C_i + k C_j$

Qr $|A| = \begin{vmatrix} 1 & -2 & 3 & 0 \\ 0 & 1 & 2 & 4 \\ -1 & 2 & 0 & -3 \\ 0 & 2 & -2 & 1 \end{vmatrix} = ? = +1 \begin{vmatrix} 1 & 2 & 4 \\ 2 & 0 & 3 \\ 2 & -2 & 1 \end{vmatrix} - 0 + (-1) \begin{vmatrix} -2 & 3 & 0 \\ 1 & 2 & 4 \\ 2 & -2 & 1 \end{vmatrix} - 0$

$\begin{vmatrix} + & - & + & - \\ - & + & - & + \\ + & - & + & - \\ - & + & - & + \end{vmatrix}$

$= \dots = \underline{\underline{-3}}$ Ans

Qr $|A| = \begin{vmatrix} -3 & 1 & 1 & 2 \\ 1 & -3 & 1 & 2 \\ 1 & 1 & -3 & 2 \\ 2 & 1 & 1 & -3 \end{vmatrix} = ?$

$C_1 \rightarrow C_1 + (C_2 + C_3 + C_4)$

$\begin{vmatrix} 1 & 1 & 1 & 2 \\ 1 & -3 & 1 & 2 \\ 1 & 1 & -3 & 2 \\ 1 & 1 & 1 & -3 \end{vmatrix}$

$\begin{matrix} R_2 - R_1 \\ R_3 - R_1 \\ R_4 - R_1 \end{matrix} \rightarrow \begin{vmatrix} 1 & 1 & 1 & 2 \\ 0 & -4 & 0 & 0 \\ 0 & 0 & -4 & 0 \\ 0 & 0 & 0 & -5 \end{vmatrix} = +1 \begin{vmatrix} -4 & 0 & 0 \\ 0 & -4 & 0 \\ 0 & 0 & -5 \end{vmatrix} = (-4)(-4)(-5) = \underline{\underline{-80}}$ Ans

The determinant value of the matrix

$$\begin{vmatrix} 13 & 2 & 1 & 3 \\ 31 & 4 & 5 & 6 \\ 26 & 3 & 7 & 4 \\ 10 & 1 & 3 & 2 \end{vmatrix} \text{ is } \underline{\hspace{2cm}}.$$

(a) 55

(b) 101

(c) 126

(d) ☒ -10

$$C_2 \rightarrow C_2 - C_4 \rightarrow \begin{vmatrix} 13 & -1 & 1 & 3 \\ 31 & -2 & 5 & 6 \\ 26 & -1 & 7 & 4 \\ 10 & -1 & 3 & 2 \end{vmatrix} \xrightarrow{\substack{R_2 - 2R_1 \\ R_3 - R_1 \\ R_4 - R_1}} \begin{vmatrix} 13 & -1 & 1 & 3 \\ 5 & 0 & 3 & 0 \\ 13 & 0 & 6 & 1 \\ -3 & 0 & 2 & -1 \end{vmatrix}$$

$$|A| = -(-1) \begin{vmatrix} 5 & 3 & 0 \\ 13 & 6 & 1 \\ -3 & 2 & -1 \end{vmatrix}$$

$$= \dots = -10$$

Properties of Determinants

$$① |ABC| = |A| \cdot |B| \cdot |C|$$

$$② |A^m| = |A|^m$$

$$③ |A^T| = |A| \Rightarrow |\text{adj} A| = |\text{Cof} A|$$

$$④ |\bar{A}| = \frac{1}{|A|}$$

$$\text{ie } \boxed{\det \bar{A} = \frac{1}{\det A}}$$

$$⑤ |A+B+C| \leq |A| + |B| + |C|$$

⑥ if $A_{n \times n}$ & K is any scalar then

$$[KA] = K[A] \text{ But } |KA| = K^n \cdot |A|$$

$$⑦ \text{Tr}(A_{n \times n}) = \sum a_{ii} \forall i$$

⑧ Det of U.T.M, L.T.M, Diagonal Mat, Scalar Mat
= Product of Diag elements

$$⑨ |\underbrace{\text{adj adj adj} \dots \text{adj}}_{n \text{ times}} A| = |A|^{(n-1)}, A_{n \times n}$$

$$\text{eg } |\text{adj} A| = |A|^{(n-1)}$$

If A and B are two matrices of order 3×5 and 5×3 respectively then determinant of the matrix $4BA$ equals

- (a) $4|B||A|$ (b) $4^3|B||A|$
 (c) $4^5|A||B|$ (d) ☒ None

$$B_{5 \times 3} A_{3 \times 5} \Rightarrow (BA)_{5 \times 5}$$

$$\therefore |4BA|_{5 \times 5} = 4^5 |BA| = \text{Correct.}$$

$$= 4^5 \cdot |B| \cdot |A| = \text{wrong}$$

$\because$ A & B are Not Sq. Matrices
 $\therefore |A| \& |B| = \text{NONE.}$

M is a squagre matrix or order n and its determinant value is 5. If all the elements of M are multiplied by 2, its determinant value becomes

40. The value of n is $n=?$

(a) 2

~~(b) 3~~

(c) 4

(d) 5

ATQ, $|M|_{n \times n} = 5$
 $\& \quad |2M|_{n \times n} = 40$
 $2^n |M| = 40$

$$2^n \cdot (5) = 40$$

$$2^n = 8$$

$$2^n = 2^3 \Rightarrow n = 3$$

If A is a square matrix of order n then $|\text{Adj}(\text{Adj} A)| = \underline{\hspace{2cm}}$.

(a) $|A|$

(b) $|A|^n$

(c) $|A|^{n-1}$

☒ (d) $|A|^{(n-1)^2}$

$$|\underbrace{\text{adj adj}}_{r=2} A| = |A|^{(n-1)^2}$$

The value of the determinant $\begin{vmatrix} -a^2 & ab & ac \\ ba & -b^2 & bc \\ ac & bc & -c^2 \end{vmatrix}$ will

be _____.

(a) $2a^2b^2c^2$

(b) $-2a^2b^2c^2$

(c) $4a^2b^2c^2$

(d) $-4a^2b^2c^2$

$$= abc \begin{vmatrix} -a & b & c \\ a & -b & c \\ a & b & -c \end{vmatrix}$$

$$= a^2b^2c^2 \begin{vmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix}$$

$$= a^2b^2c^2 \begin{vmatrix} -1 & 1 & 1 \\ 0 & 0 & 2 \\ 0 & 2 & 0 \end{vmatrix}$$

$$= a^2b^2c^2 \begin{pmatrix} -1 & 0 & -4 \end{pmatrix}$$

$$= 4a^2b^2c^2$$

$$R_2 + R_1$$

$$R_3 + R_1$$

The determinant of a matrix has 720 terms (in the unsimplified form). The order of the matrix is

— (a) 5 (b) 6 (c) 7 (d) 8

Man No. of terms in the General Expansion of $|A|_{n \times n} = ? = n!$ terms

let $A_{n \times n}$ then ATQ , $n! = 720$

$$n! = 6!$$

$$n = 6$$

Let A be a 10×10 matrix in which each row has exactly one entry to 1 the remaining nine entries of the row being 0 which of the following is not possible value for the determinant of the matrix.

(a) 0

(b) -1

(c) 10

(d) 1

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$|A| = 0, 0, 1, -1$$

$$\text{ie } |A| = -1, 0, 1$$

X and Y are non-zero square matrices of size $n \times n$. If $XY = O_{n \times n}$ then

(a) $|X| = 0$ and $|Y| \neq 0$ (b) $|X| \neq 0$ and $|Y| = 0$

(c) $|X| = 0$ and $|Y| = 0$ (d) $|X| \neq 0$ and $|Y| \neq 0$

Let $|X| \neq 0 \Rightarrow X^{-1}$ exist

So $XY = O$

$$X^{-1}(XY) = X^{-1}O$$

$$(X^{-1}X)Y = O$$

$$IY = O$$

$Y = O$ Not possible for $|X| = 0$

Similarly $|Y| \neq 0 \Rightarrow Y^{-1}$ exist

So $XY = O$

$$(XY)Y^{-1} = OY^{-1}$$

$$XI = O \Rightarrow X = O$$

Again Not possible for $|Y| = 0$

So ans = (c) ✓

Singular Mat: If $|A| = 0$ then A is called Singular Mat

Non singular Mat: If $|A| \neq 0$ then A is called Non singular Mat

Invertible Mat $\rightarrow$ If A^{-1} exist then A is called Invertible Mat

& it is defined as $A^{-1} = \frac{\text{adj } A}{|A|}$ or $\frac{(\text{Cof } A)^T}{|A|}$

N. Condition for a Matrix to be invertible is $|A| \neq 0$

Sp. Conclusion: w.k. that, $AA^{-1} = A^{-1}A = I$

so if we have,

$$AB = BA = I$$

$\Rightarrow$

Inverse of $A=B$ i.e. $A^{-1}=B$

Inverse of $B=A$ i.e. $B^{-1}=A$

i.e. Both are the Inverses of each other.

(*) If $M_{n \times n}$ s.t. $M^4 = I$ then

- $M^{-1} = ? = M^3, \because M M^3 = I$
- $(M^2)^{-1} = ? = M^2, \because M^2 \cdot M^2 = I$
- $(M^3)^{-1} = ? = M, \because M^3 \cdot M = I$

(*) Shortcut of finding Inverse of 2×2 Mat $\rightarrow$

if $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ then $A^{-1} = \frac{\text{adj } A}{|A|} = \frac{1}{|A|} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

e.g. $A = \begin{bmatrix} 2 & -6 \\ 0 & 1 \end{bmatrix}$ then $A^{-1} = ? = \frac{1}{2} \begin{bmatrix} 1 & 6 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 0.5 & 3 \\ 0 & 1 \end{bmatrix}$ Ans

$\because |A| = 2 (\neq 0 \text{ so } A^{-1} \text{ exist})$

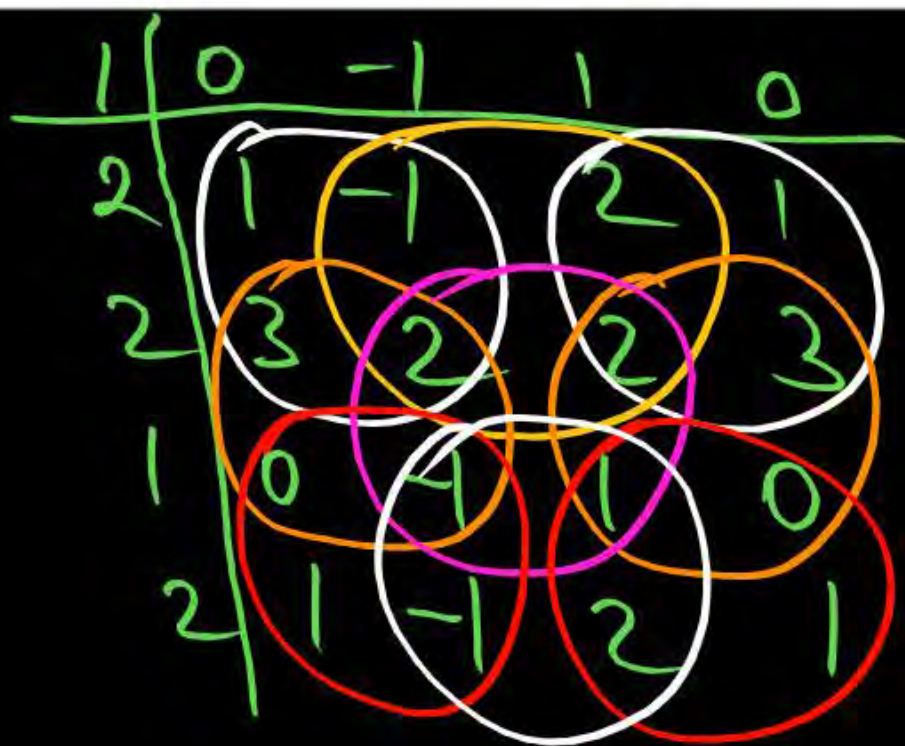
If $R = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 1 & -1 \\ 2 & 3 & 2 \end{bmatrix}$. Then top row of R^{-1} is

(a) $[5 \ 6 \ 4]$

(b) $[5 \ -3 \ 1]$

(c) $[2 \ 0 \ -1]$

(d) $\begin{bmatrix} 2 & -1 & \frac{1}{2} \end{bmatrix}$



$\therefore |R| = \dots = 1$
 $(\because |R| \neq 0 \text{ so } R^{-1} \text{ will exist})$

so $R^{-1} = \frac{\text{adj } R}{|R|} = \frac{1}{(1)} \begin{bmatrix} 5 & -3 & 1 \\ -6 & 4 & -1 \\ 4 & -3 & 1 \end{bmatrix}$ Ans

(ii) if $\text{Cof } R = \begin{bmatrix} 5 & -6 & 4 \\ -3 & 4 & -3 \\ 1 & K & 1 \end{bmatrix}$ then $K = ?$

w.k. that $\text{Cof } R = (\text{adj } A)^T = \begin{bmatrix} 5 & -6 & 4 \\ -3 & 4 & -3 \\ 1 & -1 & 1 \end{bmatrix}$
 so $K = -1$

PQ If $A = \begin{bmatrix} 1 & -2 & 3 \\ -2 & 5 & -1 \\ 3 & -9 & -6 \end{bmatrix}$ then $A^{-1} = ?$ & $\text{adj} A = ?$

$\therefore |A| = \dots = 0 \Rightarrow A^{-1} = \text{DNE}$

$\therefore R_3 = -3(R_1 + R_2)$ or $3R_1 + 3R_2 + R_3 = 0$

is R_1, R_2, R_3 are dependent $\Rightarrow |A| = 0$.

(ii)

1	-2	3	1	-2
-2	5	-1	-2	5
3	-9	-6	3	-9
1	-2	3	1	-2
-2	5	-1	-2	5

So $\text{adj} A = \begin{bmatrix} -39 & - & - \\ -15 & - & - \\ 3 & - & - \end{bmatrix}$ //

(*) Commutative Law $\begin{cases} A+B = B+A \\ AB \neq BA \text{ in general. But } \text{Tr}(AB) = \text{Tr}(BA) \text{ always.} \end{cases}$

(*) Associative Law $\begin{cases} A+(B+C) = (A+B)+C \\ A(BC) = (AB)C \end{cases}$

for eg $A_{2 \times 3}$ & $B_{3 \times 2}$ then $\begin{cases} (AB)_{2 \times 2} \text{ is } AB \neq BA \\ (BA)_{3 \times 3} \text{ But } \text{Tr}(AB) = \text{Tr}(BA) \end{cases}$

(*) Conjugate Mat $\rightarrow$ Let A is any Mat then $\begin{cases} \bar{A} = \text{Conjugate Mat of } A \\ A^T = \text{Transpose} \dots \\ A^\theta = \text{Transjugate} \dots \end{cases}$
 it is defined as $A^\theta = (\bar{A}^T)$ or $(\bar{A})^T$

(*) Real Mat $\rightarrow$ if $\boxed{\bar{A} = A}$ then A is Real Mat & $(A^\theta = A^T)$

Sp. Points:

$$\textcircled{1} \text{Tr}(A+B+C) = \text{Tr}(A) + \text{Tr}(B) + \text{Tr}(C)$$

$$\textcircled{2} (A+B+C)^{\theta} = A^{\theta} + B^{\theta} + C^{\theta}$$

$$\textcircled{3} (A+B+C)^T = A^T + B^T + C^T$$

$$\textcircled{4} |A+B+C| \leq |A| + |B| + |C|$$

$$\textcircled{5} (A+B+C)^{-1} \neq A^{-1} + B^{-1} + C^{-1}$$

But $(A+B+C)^{-1} = \frac{\text{adj}(A+B+C)}{|A+B+C|}$

$$\textcircled{6} |ABC| = |A| \cdot |B| \cdot |C|$$

REVERSAL LAW:-

$$(ABC)^T = C^T B^T A^T$$

$$(ABC)^{\theta} = C^{\theta} B^{\theta} A^{\theta}$$

$$(ABC)^{-1} = C^{-1} B^{-1} A^{-1}$$

$\otimes$ we know that, $AB \neq BA$ in general
But $\text{Tr}(AB) = \text{Tr}(BA)$ always.

If A & B are two matrices of same order then which of the following is true.

- (a) $(A + B)^2 = A^2 + 2AB + B^2$ ✗
 (b) $(A - B)^2 = A^2 - 2AB + B^2$ ✗
 (c) $(A + B)^2 + (A - B)^2 = 2A^2 + 2B^2$ (True)
 (d) $(A + B)(A - B) = A^2 - B^2$ ✗

(d)

$$(A+B)(A-B) = A^2 - AB + BA - B^2$$

$$(a) (A+B)^2 = (A+B)(A+B) = A^2 + AB + BA + B^2$$

$$(b) (A-B)^2 = (A-B)(A-B) = A^2 - AB - BA + B^2$$

If X and Y are two singular matrices such that $XY = Y$ and $YX = X$ then $X^2 + Y^2$ equals

(a) $X + Y$

(b) XY

(c) $2I$

(d) $2(X + Y)$

given, $XY = Y \rightarrow \textcircled{1}$ & $YX = X \rightarrow \textcircled{2}$

!! X & Y are Sing. Matrices so we can not apply Cancellation Law

$$\begin{aligned} X^2 + Y^2 &= XX + YY \\ &= X(YX) + Y(XY) \\ &= (XY)X + (YX)Y \\ &= YX + XY = X + Y \end{aligned}$$

For N.O. system, $x^3 = x$
 $\Rightarrow x^2 = 1 \Rightarrow x = \pm 1, (x \neq 0)$

Similarly for Matrix Algebra,

Logic: $A^2 = A \Rightarrow A = I, (|A| \neq 0)$
 Let $|A| \neq 0 \Rightarrow A^{-1}$ exist
 $\rightarrow A^{-1}(AA) = A^{-1}A$
 $IA = I \Rightarrow A = I$

The number of different matrices that can be formed with elements 0, 1, 2 and 3; each matrix having 4 elements is

(a) 2×4^4

✓ (b) 3×4^4

(c) 4×4^4

(d) 3×2^4

Possible order of $A = \begin{bmatrix} - & - \\ - & - \end{bmatrix}_{2 \times 2}$ or $\begin{bmatrix} - \\ - \\ - \\ - \end{bmatrix}_{4 \times 1}$ or $[- - - -]_{1 \times 4}$

Total ways of filling 4 elements = $\underbrace{4 \text{ ways}}_{1^{\text{st}}} \times \underbrace{4 \text{ ways}}_{2^{\text{nd}}} \times \underbrace{4 \text{ ways}}_{3^{\text{rd}}} \times \underbrace{4 \text{ ways}}_{4^{\text{th}}} = 4^4 \text{ ways}$

∴ Total Matrices = $4^4 + 4^4 + 4^4 = 3 \times 4^4$

The matrix $A = (a_{ij})_{m \times n}$ is defined as $a_{ij} = i + j$ for all i, j then the sum of all the elements of matrix

A = _____.

(a) $\frac{mn}{2}(m+n+1)$

~~(b) $\frac{mn}{2}(m+n+2)$~~

(c) $\frac{m}{2} \left(\frac{n(n+1)}{2} \right)$

(d) $\frac{n}{2} \left(\frac{m(m+1)}{2} \right)$

Let $m=2, n=3$, $A = [a_{ij}]_{2 \times 3} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} = \begin{bmatrix} 2 & 3 & 4 \\ 3 & 4 & 5 \end{bmatrix}$

Using (a): $\frac{2 \times 3}{2}(2+3+1) = 18 \neq 21$

"(b)": $\frac{2 \times 3}{2}(2+3+2) = 21 \checkmark$

$A = [a_{ij}]_{m \times n}$ $\left(a_{ij} = i + j \right)$

$\sum_{i=1}^m \sum_{j=1}^n a_{ij} = ??$

$\sum_{i=1}^2 \sum_{j=1}^3 a_{ij} = 2+3+4+3+4+5 = 21$

$\textcircled{M-Q}$ let $m=n=2$ then $A = [a_{ij}]_{2 \times 2} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix}$

$$\sum \sum a_{ij} = 2+3+3+4 = \textcircled{12}$$

Taking $\textcircled{b}$, $\frac{mn}{2}(m+n+2)$

$$= \frac{2 \times 2}{2}(2+2+2)$$

$$= 2(6)$$

$$= \textcircled{12}$$

so b is correct.

If ω be the cube root of unity then

$$\begin{bmatrix} \omega & \omega^2 \\ 1 & \omega \\ \omega^1 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & \omega & \omega^2 \\ \omega & \omega^2 & 1 \end{bmatrix} \times \begin{bmatrix} 1 \\ \omega \\ \omega^2 \end{bmatrix} \text{ equals}$$

$3 \times 2 \quad 2 \times 3 \quad 3 \times 1$

(a) $\begin{bmatrix} \omega - \omega^2 \\ \omega - \omega^2 \\ \omega - 2\omega^2 \end{bmatrix}$ (b) $\begin{bmatrix} 1 \\ \omega \\ \omega^2 \end{bmatrix}$

(c) $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} [\omega - \omega^2]$ (d) ☒ None

Note $\omega^3 = 1, 1 + \omega + \omega^2 = 0$
 $\overline{\omega} = \omega^2, \overline{\omega^2} = \omega, \frac{1}{\omega} = \omega^2$

$$A(BC) = A \begin{bmatrix} (1 + \omega^2 + \omega^1) \\ (\omega + \omega^3 + \omega^2) \end{bmatrix}_{2 \times 1}$$

$$= A \begin{bmatrix} (1 + \omega^2 + \omega) \\ (\omega + 1 + \omega) \end{bmatrix}_{2 \times 1}$$

$$= A_{3 \times 2} \begin{bmatrix} 0 \\ 0 \end{bmatrix}_{2 \times 1}$$

$$= 0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}_{3 \times 1}$$

Types of MATRICES

- ① Singular Mat $\rightarrow$ if $|A| = 0$
- ② Non Sing Mat $\rightarrow$ if $|A| \neq 0$
- ③ Invertible Mat $\rightarrow$ if A^{-1} exist.
 & N.Condⁿ is $|A| \neq 0$ & Defⁿ is
 $A^{-1} = \frac{\text{adj } A}{|A|} = \frac{(\text{cof } A)^T}{|A|}$
- ④ Symm. Mat: if $A^T = A$
- ⑤ Skew symm: if $A^T = -A$
- ⑥ Hermitian: if $A^\theta = A$
- ⑦ Skew Herm: if $A^\theta = -A$

- ⑧ Transjugate Mat: $A^\theta = (\overline{A^T})$
- ⑨ Orthogonal Mat: if $AA^T = I$
 or $(A^{-1} = A^T)$
- ⑩ Unitary Mat: if $AA^\theta = I$
 or $(A^{-1} = A^\theta)$
- ⑪ Real Mat: if $\overline{A} = A$
 or $(A^\theta = A^T)$
- ⑫ Idempotent: if $A^2 = A$
- ⑬ Involuntary: if $A^2 = I$
- ⑭ Nilpotent: if $A^k = 0$
- ⑮ U.T.M:
 if $A = [a_{ij}]$ st $a_{ij} = 0 \forall i > j$
- ⑯ L.T.M:
 if $A = [a_{ij}]$ st $a_{ij} = 0 \forall i < j$
- ⑰ Diag Mat:
 $a_{ij} = \begin{cases} 0 \\ \text{at least one element is non zero} \end{cases}, \forall i \neq j$
- ⑱ Scalar: $a_{ij} = \begin{cases} 0, & i \neq j \\ k, & i = j \end{cases}$
- ⑲ Trace(A) $= \sum a_{ii}$

eg $A = \begin{bmatrix} 2 & 3 & -5 \\ 4 & 0 & 2 \end{bmatrix}$, eg $A = \begin{bmatrix} 2 & 3+i & -5i \\ 0 & -2-i & 3-2i \end{bmatrix}$

$\bar{A} = \begin{bmatrix} 2 & 3 & -5 \\ 4 & 0 & 2 \end{bmatrix}$ then $\bar{A} = \begin{bmatrix} 2 & 3-i & +5i \\ 0 & -2+i & 3+2i \end{bmatrix}$

$\because \bar{A} = A \Rightarrow A$ is Real Mat

$$A^0 = ? = (\bar{A})^T$$

$$= \begin{bmatrix} 2 & 4 \\ 3 & 0 \\ -5 & 2 \end{bmatrix} = A^T$$

$\because \bar{A} \neq A$ so A is Complex Mat

$$\& A^0 = ? = (\bar{A})^T = \begin{bmatrix} 2 & 0 \\ 3-i & -2+i \\ 5i & 3+2i \end{bmatrix}$$

* A is an O-Mat then $AA^T = I$

$$|AA^T| = |I|$$

$$|A| \cdot |A^T| = 1$$

$$|A|^2 = 1$$

$$|A| = \pm 1$$

ie either $|A| = 1$ or $|A| = -1$

So Orthogonal Mat is always Non singular.

* Unitary Mat of Real nos
is _____ also.

$$\because A \text{ is Unitary} \Rightarrow AA^H = I \rightarrow (1)$$

$$\because A \text{ is Real also} \Rightarrow A^H = A^T \rightarrow (2)$$

$$\text{By (1) \& (2), } AA^T = I$$

So A is Orthogonal also.

$$\textcircled{x} \begin{bmatrix} 2 & -3 & 0 & 4 \\ 0 & -1 & 2 & 0 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

U.T.M

$|A| = 0$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 2 & 0 & 0 \\ 0 & 1 & 2 & 0 \\ -1 & 0 & 2 & 3 \end{bmatrix}$$

L.T.M

$|A| = 12$

$$\begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix}$$

Diag Mat

$|A| = -24$

$$\begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

Scalar Mat

$|A| = 3^4 = 81$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Identity Mat

$|A| = 1$

e.g. $|A| = \begin{vmatrix} 0 & 1 & 2 & 0 \\ -1 & 0 & 4 & 0 \\ -2 & -4 & 0 & 6 \\ 0 & 0 & -6 & 0 \end{vmatrix} = ? = \dots = \textcircled{36} \underline{\underline{Ans}}$

we can't use above property in this Mat $\because$ it is neither U.T.M, Nor L.T.M & so on...

* $\begin{bmatrix} 3 & -4 & 0 \\ -4 & 5 & 7 \\ 0 & 7 & -2 \end{bmatrix}, \begin{bmatrix} 0 & 4 & -3 \\ -4 & 0 & 1 \\ 3 & -1 & 0 \end{bmatrix}, \begin{bmatrix} 2 & 4+i & -7i \\ 4-i & -5 & 3 \\ 7i & 3 & 0 \end{bmatrix}, \begin{bmatrix} 2i & 4+i & -7i \\ -4+i & -5i & 3 \\ -7i & -3 & 0 \end{bmatrix}$

Symm Mat
($\because A^T = A$)

Skew Symm
($\because A^T = -A$)

Hermitian Mat
($\because A^\theta = A$)

Skew Hermitian
($\because A^\theta = -A$)

* Diagonal elements of Skew Symm Mat must be all Zero

* $|A|_{3 \times 3} = 0$

* If $A_{n \times n}$ s.t A is skew symm then $|A| = \begin{cases} 0, & n = \text{odd} \\ \text{perfect sq}, & n = \text{even} \end{cases}$

eg $|A| = \begin{vmatrix} 0 & -5 & 4 \\ 5 & 0 & 3 \\ -4 & -3 & 0 \end{vmatrix}_{3 \times 3}$

= skew symm of odd order
= 0

eg $|A| = \begin{vmatrix} 0 & 1 & 2 & 0 \\ -1 & 0 & 4 & 0 \\ -2 & -4 & 0 & 6 \\ 0 & 0 & -6 & 0 \end{vmatrix}_{4 \times 4}$

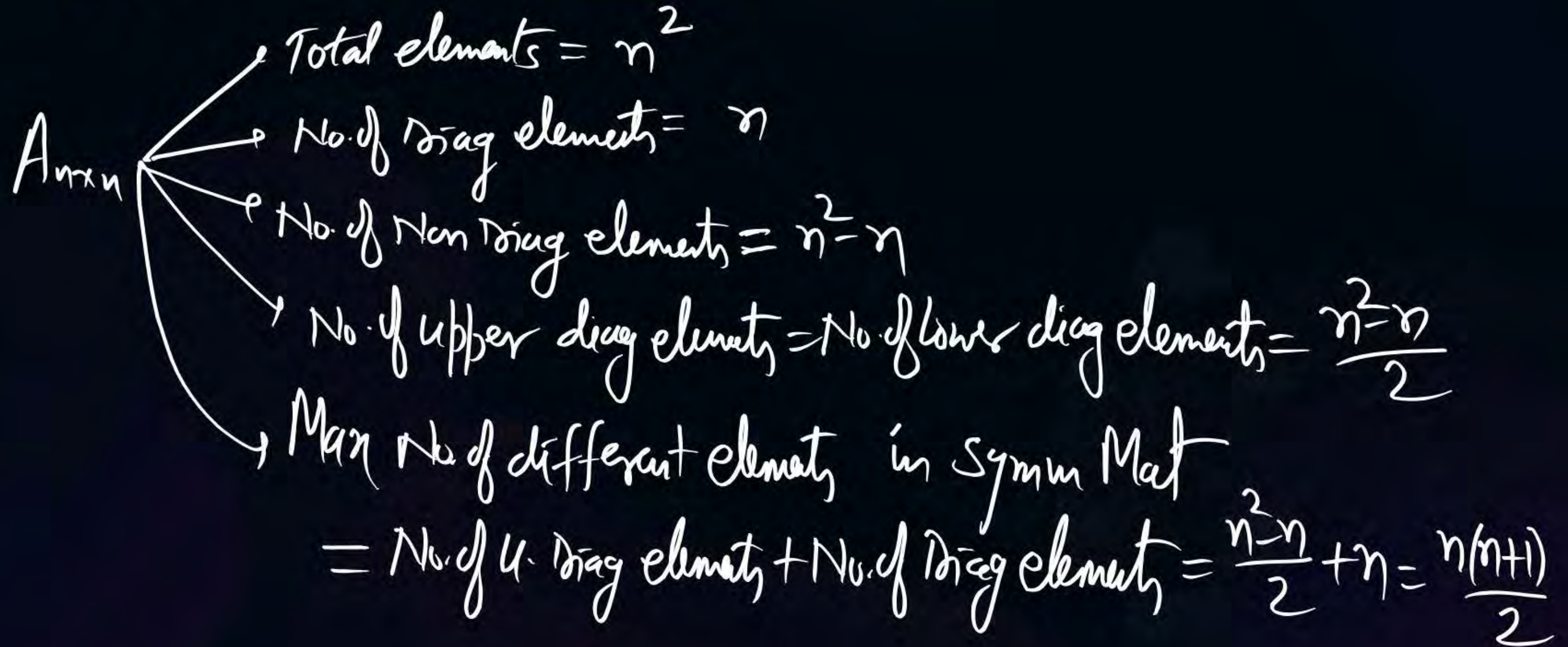
perfect sq, $n = \text{even}$
= $36 = (6)^2$
= perfect sq.

* Determinant of skew symm Mat of odd order is always ZERO.

* Take Case: Det of skew symm Mat of even order is perfect sq & may be ZERO also

$\therefore$ Zero can also be treated as perfect sq. for eg, $|A| = \begin{vmatrix} 0 & -1 & -2 & -3 \\ 1 & 0 & -1 & -2 \\ 2 & 1 & 0 & -1 \\ 3 & 2 & 1 & 0 \end{vmatrix}_{4 \times 4} = \dots = 0 = (0)^2$

* Max Number of Different elements that are required to construct a Symm Mat $A_{n \times n} = \frac{n(n+1)}{2}$



The number of different $n \times n$ symmetric matrices with each elements being either 0 or 1 is

(a) 2^n

(b) 2^{n^2}

(c) $2^{\frac{n^2+n}{2}}$

(d) $2^{\frac{n^2-n}{2}}$

Total No. of different places to fill with 0 or 1 = $\frac{n(n+1)}{2}$

So Total No. of Symm Mat = $\underbrace{2}_{1^{st}} \times \underbrace{2}_{2^{nd}} \times \underbrace{2}_{3^{rd}} \times \underbrace{2}_{4^{th}} \dots \times \underbrace{2}_{\left(\frac{n(n+1)}{2}\right)^{th}} = 2^{\frac{n(n+1)}{2}}$

* Max Number of **Ind.** entries that are required to construct skew symm Mat $A_{n \times n} = \frac{n(n-1)}{2}$

$$A = \begin{bmatrix} 0 & \alpha & \beta \\ -\alpha & 0 & \gamma \\ -\beta & -\gamma & 0 \end{bmatrix}_{3 \times 3} \quad \text{Total Ind entries} = ? = 3$$

$$A_{n \times n} \begin{cases} \text{Total entries} = n^2 \\ \text{Diag entries} = n \quad (\text{these are dependent entries}) \\ \text{Upper diag entries} = \frac{n^2 - n}{2} \end{cases}$$

If these are Ind entries, so $A_n = \frac{n^2 - n}{2} = \frac{n(n-1)}{2}$

Q If the Mat $A = \begin{bmatrix} 6 & 9 \\ -4 & -6 \end{bmatrix}$ will be ?



Let us calculate, $A^2 = AA = \begin{bmatrix} 6 & 9 \\ -4 & -6 \end{bmatrix} \begin{bmatrix} 6 & 9 \\ -4 & -6 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O$

$\therefore A^2 = O \Rightarrow A$ is Nilpotent Mat of index '2'

If T is Idempotent then $T^k = T$ for

- (a) $k = 2$
- (b) All integer k $\times$
- (c) ☒ All positive integer $k \geq 2$
- (d) All of the above

ie $T^2 = T^3 = T^4 = T^5 = \dots = T$
 $T^k = T$
 $\forall k \geq 2$

T is an Idempotent then $T^2 = T$

But

$$T^3 = T^2 \cdot T = T \cdot T = T^2 = T$$

$$\& T^4 = T^2 \cdot T^2 = T \cdot T = T^2 = T$$

Q2 HW if $A = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix}$ then A is

(a) orthogonal

(b) ☒ Involutary $\because A^2 = I$

(c) ☒ unitary $\because AA^H = I$

(d) ☒ Hermitian $\because A^H = A$

For the given orthogonal matrix Q.

$$Q = \begin{bmatrix} 3/7 & 2/7 & 6/7 \\ -6/7 & 3/7 & 2/7 \\ 2/7 & 6/7 & -3/7 \end{bmatrix}$$

then $Q^{-1} = ? = Q^T = \text{C}$ ✓

The inverse is

(a) $\begin{bmatrix} 3/7 & 2/7 & 6/7 \\ -6/7 & 3/7 & 2/7 \\ 2/7 & 6/7 & -3/7 \end{bmatrix}$

(b) $\begin{bmatrix} -3/7 & -2/7 & -6/7 \\ 6/7 & -3/7 & -2/7 \\ -2/7 & -6/7 & 3/7 \end{bmatrix}$

(c) $\begin{bmatrix} 3/7 & -6/7 & 2/7 \\ 2/7 & 3/7 & 6/7 \\ 6/7 & 2/7 & -3/7 \end{bmatrix}$

(d) $\begin{bmatrix} -3/7 & -6/7 & -2/7 \\ -2/7 & -3/7 & -6/7 \\ -6/7 & -2/7 & 3/7 \end{bmatrix}$



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THANK
you